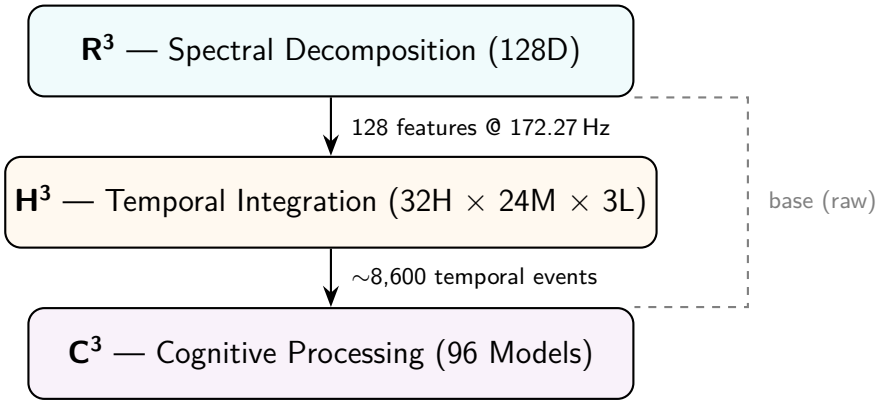
Universal Spectral Language

Terminology Reference

L^3 — Language Layer



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Musical Intelligence System

One system. One language. Spectral.

This document defines the complete terminological vocabulary used across all layers of the MI system. Every name, every address, every reference follows the conventions defined herein.

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Core Principle

R^3 names describe **what is measured** (spectral property).
H^3 names describe **how it evolves** (temporal morphology).
C^3 models **interpret** spectral evidence (emergent cognition).

No music-theory term appears as a computational primitive in R^3 or H^3.
"Chord", "key", "cadence" are C^3-emergent concepts.

R^3 — Spectral Vocabulary (128 Features)

The complete R^3 spectral representation comprises 128 normalised features organised into 11 groups (A–K). Each feature is computed per analysis frame and mapped to $[0, 1]$. The *Legacy Name* column shows the original internal code identifier; the *USL Name* column shows the standardised Universal Spectral Language identifier used throughout this document.

Table 1: Complete R^3 feature index (128 dimensions).

Idx	Group	Legacy Name	USL Name	Spectral Definition	Range
Group A: Spectral Consonance [0:6]					
0	A	roughness	roughness	Plomp–Levelt critical band interaction	$[0, 1]$
1	A	sethares_dissonanc	sethares_dissonance	Partial-pair dissonance sum (Sethares 1993)	$[0, 1]$
2	A	helmholtz_kang	helmholtz_fusion	Harmonic series alignment index	$[0, 1]$
3	A	stumpf_fusion	stumpf_fusion	Perceptual fusion of simultaneous partials	$[0, 1]$
4	A	sensory_pleasantne	sensory_pleasantnes	Composite spectral smoothness measure	$[0, 1]$
5	A	inharmonicity	inharmonicity	Deviation of partials from harmonic series	$[0, 1]$
6	A	harmonic_deviation	partial_deviation	Weighted partial frequency deviation	$[0, 1]$
Group B: Spectral Energy [7:11]					
7	B	amplitude	amplitude	RMS spectral energy	$[0, 1]$
8	B	velocity_A	amplitude_velocity	$d(\text{amplitude})/dt$ first difference	$[0, 1]$
9	B	acceleration_A	amplitude_accelerat	$d^2(\text{amplitude})/dt^2$ second difference	$[0, 1]$
10	B	loudness	loudness	Perceptual loudness (Stevens)	$[0, 1]$
11	B	onset_strength	onset_strength	Spectral flux thresholded for onset detection	$[0, 1]$
Group C: Spectral Timbre [12:20]					
12	C	warmth	spectral_warmth	Low-frequency spectral energy ratio	$[0, 1]$
13	C	sharpness	spectral_sharpness	High-frequency weighted centroid (Zwicker)	$[0, 1]$
14	C	tonalness	spectral_periodicit	Degree of harmonic structure in spectrum	$[0, 1]$
15	C	clarity	spectral_clarity	Peak-to-average spectral energy ratio	$[0, 1]$
16	C	spectral_smoothnes	spectral_smoothness	Inverse spectral irregularity	$[0, 1]$
17	C	spectral_autocorre	spectral_autocorrel	Spectral self-similarity	$[0, 1]$
18	C	tristimulus1	tristimulus_low	Fundamental partial energy ratio	$[0, 1]$
19	C	tristimulus2	tristimulus_mid	Partials 2–4 energy ratio	$[0, 1]$
20	C	tristimulus3	tristimulus_high	Partials 5+ energy ratio	$[0, 1]$
Group D: Spectral Change [21:24]					
21	D	spectral_flux	spectral_flux	Frame-to-frame L^2 spectral distance	$[0, 1]$
22	D	distribution_entro	spectral_entropy	Shannon entropy of spectral distribution	$[0, 1]$

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Idx	Group	Legacy Name	USL Name	Spectral Definition	Range
23	D	distribution_flatn	spectral_flatness	Geometric/arithmetic mean ratio of spectrum	[0, 1]
24	D	distribution_conce	spectral_concentrat	Spectral energy concentration measure	[0, 1]
Group E: Spectral Interactions [25:48]					
25–32	E	x_l0l5_0..7	energy_consonance_0	Energy × Consonance cross-product (8 dims)	[0, 1]
33–40	E	x_l4l5_0..7	change_consonance_0	Change × Consonance cross-product (8 dims)	[0, 1]
41–48	E	x_l5l7_0..7	consonance_timbre_0	Consonance × Timbre cross-product (8 dims)	[0, 1]
Group F: Pitch-Class Distribution [49:64]					
49	F	chroma_C	pce_0	Spectral energy in pitch class 0 (C)	[0, 1]
50	F	chroma_Db	pce_1	Spectral energy in pitch class 1 (C♯/D♭)	[0, 1]
51	F	chroma_D	pce_2	Spectral energy in pitch class 2 (D)	[0, 1]
52	F	chroma_Eb	pce_3	Spectral energy in pitch class 3 (D♯/E♭)	[0, 1]
53	F	chroma_E	pce_4	Spectral energy in pitch class 4 (E)	[0, 1]
54	F	chroma_F	pce_5	Spectral energy in pitch class 5 (F)	[0, 1]
55	F	chroma_Gb	pce_6	Spectral energy in pitch class 6 (F♯/G♭)	[0, 1]
56	F	chroma_G	pce_7	Spectral energy in pitch class 7 (G)	[0, 1]
57	F	chroma_Ab	pce_8	Spectral energy in pitch class 8 (G♯/A♭)	[0, 1]
58	F	chroma_A	pce_9	Spectral energy in pitch class 9 (A)	[0, 1]
59	F	chroma_Bb	pce_10	Spectral energy in pitch class 10 (A♯/B♭)	[0, 1]
60	F	chroma_B	pce_11	Spectral energy in pitch class 11 (B)	[0, 1]
61	F	pitch_height	spectral_centroid_l	$\log_2$-weighted spectral centroid	[0, 1]
62	F	pitch_class_entrop	pce_entropy	Shannon entropy of 12D PCE distribution	[0, 1]
63	F	pitch_salience	pce_salience	Peak-to-median ratio in PCE distribution	[0, 1]
64	F	inharmonicity_inde	spectral_inharmonic	1 − (PCE peak/PCE sum)	[0, 1]
Group G: Onset Periodicity [65:74]					
65	G	tempo_estimate	onset_period	Dominant period in onset autocorrelation	[0, 1]
66	G	beat_strength	onset_autocorr_peak	Autocorrelation value at dominant period	[0, 1]
67	G	pulse_clarity	onset_periodicity_c	Dominant peak / median ratio in onset autocorr	[0, 1]
68	G	syncopation_index	onset_offbeat_ratio	Onset energy at off-period positions	[0, 1]
69	G	metricality_index	onset_subdivision_d	Count of significant nested subdivisions	[0, 1]
70	G	isochrony_nPVI	onset_interval_regu	1 − nPVI of inter-onset intervals	[0, 1]
71	G	groove_index	onset_bass_pulse_pr	offbeat × bass_energy × periodicity_clarity	[0, 1]
72	G	event_density	onset_density	Onset count per second (normalised)	[0, 1]
73	G	tempo_stability	onset_period_stabil	1 − CV of local onset periods	[0, 1]
74	G	rhythmic_regularit	onset_interval_entr	1− normalised IOI histogram entropy	[0, 1]
Group H: Spectral Harmony [75:86]					
75	H	key_clarity	pcp_correlation_peal	Max correlation with 24 K–K profiles	[0, 1]
76	H	tonnetz_fifth_x	pcg_fifth_x	Pitch-class geometry: fifths circle cos	[0, 1]
77	H	tonnetz_fifth_y	pcg_fifth_y	Pitch-class geometry: fifths circle sin	[0, 1]
78	H	tonnetz_minor_x	pcg_minor_x	Pitch-class geometry: minor-3rd circle cos	[0, 1]

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Table 1 continued from previous page

Idx	Group	Legacy Name	USL Name	Spectral Definition	Range
79	H	tonnetz_minor_y	pcg_minor_y	Pitch-class geometry: minor-3rd circle sin	$[0, 1]$
80	H	tonnetz_major_x	pcg_major_x	Pitch-class geometry: major-3rd circle cos	$[0, 1]$
81	H	tonnetz_major_y	pcg_major_y	Pitch-class geometry: major-3rd circle sin	$[0, 1]$
82	H	voice_leading_dist	chroma_l1_distance	L^1 norm of successive PCE differences / 2	$[0, 1]$
83	H	harmonic_change	chroma_frame_distan	1− cosine similarity of successive PCE	$[0, 1]$
84	H	tonal_stability	chroma_temporal_con	pcp_corr_peak × (1− smoothed chroma_dist)	$[0, 1]$
85	H	diatonicity	pcp_template_fit	Max fit to diatonic pitch-class template	$[0, 1]$
86	H	syntactic_irregula	chroma_reference_di	KL divergence from best-fit reference profile	$[0, 1]$
Group I: Spectral Information [87:93]					
87	I	melodic_entropy	pce_transition_entr	Entropy of pitch-class transition distribution	$[0, 1]$
88	I	harmonic_entropy	chroma_distribution_	$\mathrm{KL}(\mathrm{PCE}_t \,\|\, \mathrm{PCE}_{\mathrm{avg}})$ exponential mapping	$[0, 1]$
89	I	rhythmic_informati	onset_interval_surp	$-\log(p(\mathrm{IOI}))$ from running IOI histogram	$[0, 1]$
90	I	spectral_surprise	spectral_distributi	$\mathrm{KL}(\mathrm{mel}_t \,\|\, \mathrm{mel}_{\mathrm{avg}})$ exponential mapping	$[0, 1]$
91	I	information_rate	spectral_mutual_inf	$\mathrm{MI}(\mathrm{mel}_t; \mathrm{mel}_{t-1})/\log(128)$	$[0, 1]$
92	I	predictive_entropy	spectral_prediction_	$0.5 \cdot \log(2\pi e \cdot \sigma^2_{\mathrm{res}})/\log(128)$	$[0, 1]$
93	I	tonal_ambiguity	pce_keyed_entropy	Entropy of softmax(K–K corr × 5)	$[0, 1]$
Group J: Spectral Timbre Extended [94:113]					
94–106	J	mfcc_1 .. 13	mfcc_1 .. mfcc_13	DCT-II coefficients of log-mel spectrum (13 dims)	$[0, 1]$
107–113	J	spectral_contrast_	spectral_contrast_1	Peak–valley contrast in 7 octave sub-bands	$[0, 1]$
Group K: Spectral Modulation [114:127]					
114	K	modulation_0_5Hz	mod_energy_0.5hz	Temporal modulation energy at 0.5 Hz	$[0, 1]$
115	K	modulation_1Hz	mod_energy_1hz	Temporal modulation energy at 1 Hz	$[0, 1]$
116	K	modulation_2Hz	mod_energy_2hz	Temporal modulation energy at 2 Hz	$[0, 1]$
117	K	modulation_4Hz	mod_energy_4hz	Temporal modulation energy at 4 Hz	$[0, 1]$
118	K	modulation_8Hz	mod_energy_8hz	Temporal modulation energy at 8 Hz	$[0, 1]$
119	K	modulation_16Hz	mod_energy_16hz	Temporal modulation energy at 16 Hz	$[0, 1]$
120	K	modulation_centroi	mod_centroid	Centroid of modulation spectrum	$[0, 1]$
121	K	modulation_bandwid	mod_bandwidth	Bandwidth of modulation spectrum	$[0, 1]$
122	K	sharpness_zwicker	sharpness_zwicker	Zwicker sharpness (Bark-weighted)	$[0, 1]$
123	K	fluctuation_streng	fluctuation_strengt	Fluctuation strength (4 Hz peak)	$[0, 1]$
124	K	loudness_a_weighte	loudness_a_weighted	A-weighted loudness	$[0, 1]$
125	K	alpha_ratio	alpha_ratio	Energy ratio below/above 1 kHz	$[0, 1]$
126	K	hammarberg_index	hammarberg_index	Energy peak ratio 0–2 kHz / 2–5 kHz	$[0, 1]$
127	K	spectral_slope_0_5	spectral_slope_0_50	Linear regression slope of spectrum 0–500 Hz	$[0, 1]$

H^3 Temporal Demand Space

$R^3 \to H^3$ Dimensional Expansion

R^3 produces 128 features per analysis frame at 172.27 Hz. Each feature $r_i[t]$ is a *time series*: a sequence of scalar values across T frames. H^3 takes these time series and applies **temporal morphology**—computing how each spectral property *behaves over time* at multiple temporal scales.

The four-dimensional demand. Every H^3 output is defined by a 4-tuple:

Component	What it selects	Size
`r3_idx`	Which R^3 feature to observe	128
`horizon`	How wide a temporal window to use	32
`morph`	Which statistical descriptor to compute	24
`law`	Which temporal perspective (causal direction)	3

Theoretical vs. active space. The full Cartesian product yields $128 \times 32 \times 24 \times 3 =$ **294,912** possible temporal demands. In practice, each C^3 model declares only the demands it needs, yielding **∼8,600 active demands** across all 96 models.

Computation. For a given demand (i, h, m, ℓ) at frame t:

1. **Select** the R^3 feature series: $r_i[t]$.
2. **Window** using horizon h: extract n_h frames according to law ℓ:
   - $\ell = \text{L0}$ (memory): $[t - n_h + 1,\ t]$ — past → present
   - $\ell = \text{L1}$ (prediction): $[t,\ t + n_h - 1]$ — present → future
   - $\ell = \text{L2}$ (integration): $[t - n_h/2,\ t + n_h/2]$ — symmetric
3. **Weight** the window with exponential kernel: $A(\Delta t) = \exp(-3|\Delta t|/H)$.
4. **Apply morph** m to the weighted window → single scalar output.

Output shape. Each active demand produces one scalar per frame: the "temporal morphology" of that spectral feature at that time scale. The complete H^3 output for a given frame is the vector of all active demand values.

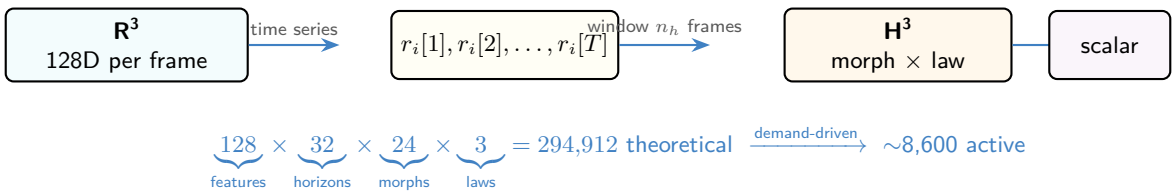

Figure 1: $R^3 \to H^3$ dimensional expansion. Each R^3 feature becomes a time series; H^3 applies temporal morphology (horizon × morph × law) to produce scalar temporal descriptors. C^3 models select only the demands they need from the 294,912 theoretical space.

Concrete example. Consider C^3 model EDNR (Expectancy-Driven Neural Response, PCU) requesting the demand `83.H12.M9.L0`:

Component	Value	Meaning
`r3_idx = 83`	`chroma_frame_distance`	Cosine distance between successive chroma vectors
`horizon = H12`	`beat` (525 ms, 90 frames)	Standard beat-level temporal scale
`morph = M9`	`velocity_mean`	Mean rate of change over the window
`law = L0`	`memory` (causal)	Past → present: how fast *was* chroma changing

Result: a single scalar per frame telling EDNR "the average speed at which spectral pitch-class distribution changed over the last beat."

H³ Output Classification: Named Temporal Properties

R³ features are raw spectral measurements. After H³ processing, each of the ∼8,600 active measurements acquires a **named temporal identity** at three levels:

Level	What it names	Count	Example
1. Class	R³ Group × Morph Category	55	*Consonance Motion*
2. Type	Class + specific feature + specific morph	3,072*	*roughness velocity*
3. Instance	Type + horizon + law	∼8,600†	`roughness.velocity.beat.memory`

*128 features × 24 morphs = 3,072 theoretical types. †Demand-driven subset of 294,912.

The 55 **classes** (Table 2) are the top-level semantic taxonomy. Each class contains many individual measurements. For example, the class "Consonance Level" spans 7 features × 8 distribution morphs × up to 32 horizons × 3 laws.

Table 2: Level-1 taxonomy: 55 named temporal property classes (11 R³ groups × 5 morph categories). Each cell is a *class name* containing many individual measurements. Horizon and law refine each class into specific Level-3 instances (e.g. "Consonance Motion" → `roughness.velocity.beat.memory`).

R³ Group	**Distribution** M0–M7	**Dynamics** M8–M13,15,18,21	**Periodicity** M14,17,22	**Information** M20	**Geometry** M16,19,23
A Consonance [0:6]	Consonance Level	Consonance Motion	Consonance Periodicity	Consonance Predictability	Consonance Contour
B Energy [7:11]	Energy Level	Energy Momentum	Energy Pulse	Energy Predictability	Energy Contour
C Timbre [12:20]	Timbre Profile	Timbre Drift	Timbre Periodicity	Timbre Predictability	Timbre Contour
D Change [21:24]	Change Level	Change Acceleration	Change Periodicity	Change Predictability	Change Contour
E Interactions [25:48]	Interaction Level	Interaction Dynamics	Interaction Periodicity	Interaction Predictability	Interaction Contour
F Pitch-Class [49:64]	Pitch-Class Profile	Pitch-Class Motion	Pitch-Class Periodicity	Pitch-Class Predictability	Pitch-Class Contour
G Onset Period [65:74]	Onset Stability	Onset Momentum	Onset Regularity	Onset Predictability	Onset Contour
H Harmony [75:86]	Harmonic Stability	Harmonic Motion	Harmonic Periodicity	Harmonic Predictability	Harmonic Contour
I Information [87:93]	Info Level	Info Dynamics	Info Periodicity	Info Meta-Predictability	Info Contour
J Timbre Ext. [94:113]	Texture Profile	Texture Drift	Texture Periodicity	Texture Predictability	Texture Contour
K Modulation [114:127]	Modulation Level	Modulation Dynamics	Modulation Periodicity	Modulation Predictability	Modulation Contour

Reading the matrix. Each cell is a **named temporal property class**. Adding a horizon and law makes it fully specific. Examples:

Property Class	Full Address	C^3 reads as
Consonance Level	`0.H12.M1.L0`	Mean roughness over the last beat (memory)
Consonance Motion	`0.H12.M8.L0`	How fast roughness is changing at beat scale
Harmonic Stability	`75.H16.M2.L2`	Std. dev. of key clarity over one measure (integrated)
Harmonic Periodicity	`83.H20.M14.L2`	Is chroma distance periodic at phrase scale?
Energy Momentum	`10.H12.M9.L0`	Mean velocity of loudness at beat scale (memory)
Onset Regularity	`70.H18.M14.L2`	Is onset interval regularity periodic over two measures?
Info Meta-Predictability	`92.H22.M20.L0`	Entropy of predictive entropy over a section

The naming rule. Every H^3 output name follows:

H^3 Temporal Property = R^3 Group Name + Morph Category Name + Horizon + Law

`Consonance.Motion.beat.memory` = `0.H12.M8.L0`

R^3 decomposes → H^3 names → C^3 interprets.

H^3 Horizon Table (32 Temporal Scales)

Index	USL Label	Duration	Frames	Band	Spectral Scale
H0	`micro_0`	5.8 ms	1	Micro	Sub-neural integration
H1	`micro_1`	11.6 ms	2	Micro	Neural integration
H2	`micro_2`	17.4 ms	3	Micro	Onset detection boundary
H3	`micro_3`	23.2 ms	4	Micro	Phoneme boundary
H4	`micro_4`	34.8 ms	6	Micro	Pitch period scale
H5	`micro_5`	46.4 ms	8	Micro	Pitch integration scale
H6	`sub_beat`	200 ms	34	Micro	Sub-beat / note onset
H7	`short_note`	250 ms	43	Micro	Short note duration
H8	`eighth`	300 ms	52	Meso	Eighth-note scale
H9	`fast_beat`	350 ms	60	Meso	Fast beat (170 BPM)
H10	`mod_beat`	400 ms	69	Meso	Moderate beat (150 BPM)
H11	`beat_low`	450 ms	78	Meso	Lower beat (130 BPM)
H12	`beat`	525 ms	90	Meso	Standard beat (115 BPM)
H13	`beat_slow`	600 ms	103	Meso	Slow beat (100 BPM)
H14	`dotted`	700 ms	121	Meso	Dotted quarter scale
H15	`two_beats`	800 ms	138	Meso	Two-beat grouping
H16	`measure`	1,000 ms	172	Macro	Single measure scale
H17	`measure_1h`	1,500 ms	259	Macro	1.5 measure scale
H18	`two_meas`	2,000 ms	345	Macro	Two-measure scale
H19	`three_meas`	3,000 ms	517	Macro	Three-measure scale
H20	`phrase_s`	5,000 ms	861	Macro	Short phrase (5 s)
H21	`phrase_l`	8,000 ms	1,378	Macro	Long phrase (8 s)
H22	`section`	15,000 ms	2,584	Macro	Section (15 s)
H23	`section_l`	25,000 ms	4,307	Macro	Extended section (25 s)
H24	`movement_s`	36 s	6,202	Ultra	Short movement
H25	`one_min`	60 s	10,336	Ultra	One minute
H26	`two_min`	120 s	20,672	Ultra	Two minutes
H27	`movement_m`	200 s	34,453	Ultra	Medium movement
H28	`movement_l`	414 s	71,319	Ultra	Standard movement
H29	`movement_x`	600 s	103,359	Ultra	Long movement
H30	`form`	800 s	137,812	Ultra	Extended form
H31	`full_work`	981 s	168,999	Ultra	Complete work

H^3 Morph Table (24 Temporal Descriptors)

Index	USL Name	Formula	Category	Min Frames	Output Range
M0	`attn_mean`	$\sum(A \cdot x)/\sum(A)$	Distribution	1	$[0, 1]$
M1	`mean`	$\sum(x)/N$	Distribution	1	$[0, 1]$
M2	`std`	$\sqrt{\mathrm{var}(x)}$	Distribution	2	$[0, \infty)$
M3	`median`	middle value	Distribution	1	$[0, 1]$
M4	`max`	$\max(x)$	Distribution	1	$[0, 1]$
M5	`range`	$\max(x) - \min(x)$	Distribution	2	$[0, 1]$
M6	`skewness`	μ_3/σ^3	Distribution	3	$(-\infty, \infty)$
M7	`kurtosis`	μ_4/σ^4	Distribution	4	$[1, \infty)$
M8	`velocity`	dx/dt	Dynamics	2	$(-\infty, \infty)$
M9	`velocity_mean`	$\mathrm{mean}(dx/dt)$	Dynamics	2	$(-\infty, \infty)$
M10	`velocity_std`	$\mathrm{std}(dx/dt)$	Dynamics	2	$[0, \infty)$
M11	`acceleration`	d^2x/dt^2	Dynamics	3	$(-\infty, \infty)$
M12	`accel_mean`	$\mathrm{mean}(d^2x/dt^2)$	Dynamics	3	$(-\infty, \infty)$
M13	`accel_std`	$\mathrm{std}(d^2x/dt^2)$	Dynamics	3	$[0, \infty)$
M14	`periodicity`	$\max(\mathrm{autocorr}[\mathrm{lag} > 0])$	Rhythm	4	$[0, 1]$
M15	`smoothness`	$1/(1 + \lvert\mathrm{jerk}\rvert/\sigma)$	Dynamics	3	$(0, 1]$
M16	`curvature`	$\mathrm{mean}(\lvert d^2x/dt^2\rvert)$	Symmetry	3	$[0, \infty)$
M17	`shape_period`	velocity zero-crossing Δ	Rhythm	4	$[0, \infty)$
M18	`trend`	OLS regression slope	Dynamics	2	$(-\infty, \infty)$
M19	`stability`	$1/(1 + \mathrm{var}/\sigma^2)$	Symmetry	2	$(0, 1]$
M20	`entropy`	$-\Sigma\, p \cdot \log(p)\ /\ \log(16)$	Information	4	$[0, 1]$
M21	`zero_cross`	sign-change count	Dynamics	2	$[0, N-1]$
M22	`peaks`	local maxima count	Rhythm	3	$[0, N-2]$
M23	`symmetry`	$\mathrm{corr}(x, \mathrm{reverse}(x))$	Symmetry	2	$[-1, 1]$

H³ Law Table (3 Temporal Perspectives)

Index	USL Name	Direction	Kernel	Window
L0	`memory`	Past → Present	$A(dt) = \exp(-3\lvert dt\rvert/H)$, causal	$[t-n+1,\ t]$
L1	`prediction`	Present → Future	$A(dt) = \exp(-3\lvert dt\rvert/H)$, anti-causal	$[t,\ t+n-1]$
L2	`integration`	Past ↔ Future	$A(dt) = \exp(-3\lvert dt\rvert/H)$, symmetric	$[t-n/2,\ t+n/2]$

Universal Spectral Address System

Every datum flowing through the MI pipeline carries a *spectral address*: a four-component tuple that uniquely identifies its origin in the R^3–H^3 feature–temporal lattice.

```
{r3_idx}.{horizon}.{morph}.{law}
```

Numeric example: `83.H12.M9.L0`
Spectral label: `chroma_frame_distance.beat.velocity_mean.memory`

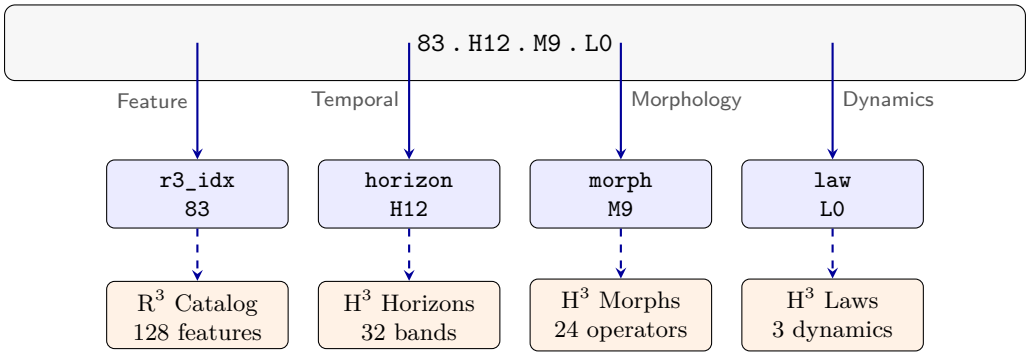

Figure 2: Four-tuple spectral address decomposition. Each component maps to a specific catalog layer in the R^3/H^3 system.

Address Construction Rules

Table 5: Address component construction rules.

Component	Source	Range	Example
`r3_idx`	R^3 Feature Index	0–127	83
`horizon`	H^3 Horizon Index	H0–H31	H12
`morph`	H^3 Morph Index	M0–M23	M9
`law`	H^3 Law Index	L0–L2	L0

Spectral Label Construction

Table 6: Spectral label construction from address components.

Component	Format	Example
Feature	R^3 USL name	`chroma_frame_distance`
Horizon	H^3 USL label	`beat`
Morph	H^3 morph name	`velocity_mean`
Law	H^3 law name	`memory`
Full label	`feature.horizon.morph.law`	`chroma_frame_distance.beat.velocity_mean.memory`

C^3 Unit Interface Table

Table 7 lists all nine C^3 processing units and their primary interfaces with the R^3 feature groups and H^3 temporal bands. R^3 groups are identified by letter (A = Consonance, B = Energy, C = Timbre, D = Change, E = Interactions, F = Pitch-Class, G = Onset Period, H = Harmony, I = Information, J = Timbre Ext., K = Modulation). H^3 bands span Micro (<50 ms), Meso (50 ms–2 s), and Macro (>2 s) perceptual time scales.

Table 7: C^3 unit interface summary.

Unit	Full Name	Models	Primary R^3 Groups	Primary H^3 Bands	Primary Laws
SPU	Sensory Processing	9	A, B, C, D	Micro–Meso	L0, L2
STU	Structural Timing	20	G, H, I	Meso–Macro	L0, L1, L2
IMU	Integrative Memory	10	F, H, I	Meso–Macro	L0, L2
ASU	Auditory Scene	9	A, B, C	Micro–Meso	L2
NDU	Neuromodulation	8	B, H, I	Meso–Macro	L0, L1, L2
MPU	Motor Planning	11	G, H	Meso	L0, L1
PCU	Predictive Coding	10	F, H, I	Micro–Macro	L0, L1, L2
ARU	Arousal Regulation	9	B, H, I	Meso–Macro	L0, L1, L2
RPU	Reward Processing	10	H, I, K	Meso–Macro	L0, L1, L2

Provenance Chain

Figure 3 traces the complete data-flow from raw audio to the C^3 input package. Every tensor element retains a spectral address linking it back to its R^3 source through an unbroken provenance chain.

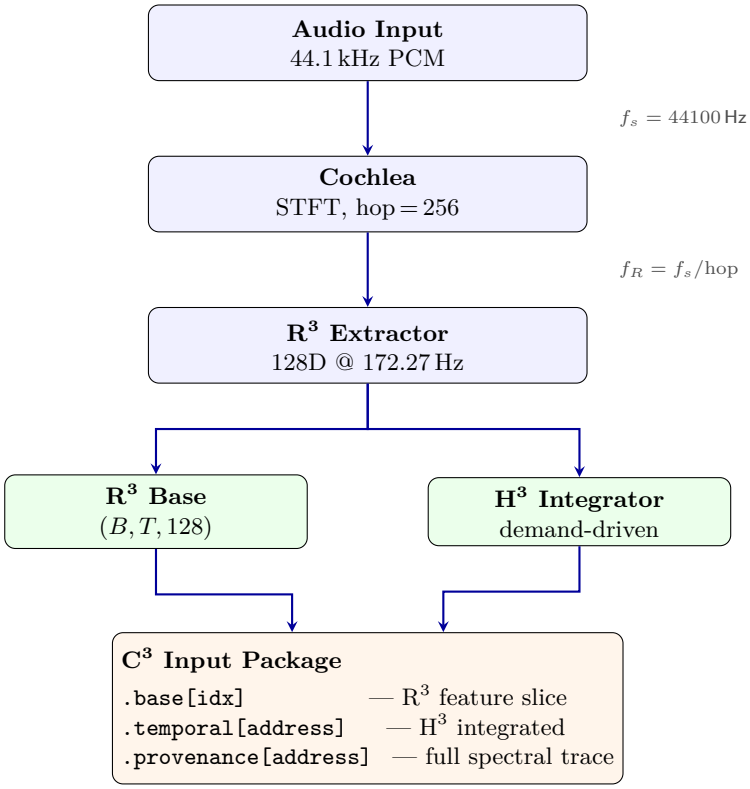

Figure 3: Provenance chain from audio input to C^3 input package. Every tensor element traces back to its R^3 source through spectral addresses.

Naming Convention Rules

The following rules govern all naming within the Universal Spectral Language. Violations break the provenance chain and are treated as specification errors.

1. **Measurement, not interpretation.** R^3 feature names describe what is *measured* (e.g. `spectral_centroid`), never what is *perceived* or *interpreted*.

2. **No music-theory terms in R^3/H^3.** Names must not contain `chord`, `key`, `cadence`, `tonal`, or `harmonic` in their music-theoretic senses. `harmonic` is permitted only in its signal-processing meaning (integer-multiple partials).

3. **Perceptual time scales, not musical structure.** H^3 horizon labels denote perceptual durations. For example, `beat` designates the $\sim$525 ms periodicity band, not a musical beat or metrical position.

4. **Dot-separated addresses.** Full spectral addresses use dot separation: `feature.horizon.morph.law`. No spaces, no slashes, no camelCase.

5. **C^3 music-theory mapping.** C^3 model documentation *may* reference music-theory terms for explanatory purposes, but every such reference *must* map to one or more spectral addresses.

6. **Unbroken spectral chain.** Every C^3 input must trace back to an R^3 source through the address system. No datum enters C^3 without a provenance record.